Name

Section - 32

Answer to the question no. 1

Given,

$f(x,y) = x^2y + y^2$ (1,3) point

Here,

$f_x = 2xy$

$f_y = x^2 + 2y$

$f_{xx} = 2y$

$f_{yy} = 2$

$f_{xy} = 2x$

Now,

$f(1,3) = 12$

$f_x(1,3) = 6$

$f_y(1,3) = 7$

$f_{xx}(1,3) = 6$

$f_{yy}(1,3) = 2$

$f_{xy}(1,3) = 2$

1st Degree Taylor Polynomial:

$$L(x,y) = f(1,3) + f_x(1,3)(x-1) + f_y(1,3)(y-3)$$

$$= 12 + 6(x-1) + 7(y-3)$$

$$= 12 + 6x - 6 + 7y - 21$$

$$= 6x + 7y - 15$$

2nd Degree Taylor Polynomial:

$$Q(x,y) = L(x,y) + \frac{f_{xx}(1,3)}{2}(x-1)^2 + f_{xy}(1,3)(x-1)(y-3) + \frac{f_{yy}(1,3)}{2}(y-3)^2$$

$$= 6x + 7y - 15 + 3(x-1)^2 + 2(x-1)(y-3) + (y-3)^2$$

$$= 6x + 7y - 15 + 3(x^2 - 2x + 1) + (2x-2)(y-3) + y^2 - 6y + 9$$

$= 6x + 7y - 15 + 3x^2 - 6x + 1 - 6x + 3 + 2xy - 6x - 2y + 6 + y^2 - 6y + 9$

$= -y + 3x^2 + 3 + 2xy + y^2 - 6x$

$= 3x^2 + y^2 + 2xy - 6x - y + 3$ (Ans.)

Answer to the question no. 2

Given,

$f(x,y) = \ln(x^2+y^2+1)$ at $(0,0)$ point

Here,

$f_x = \frac{2x}{x^2+y^2+1}$

$f_y = \frac{2y}{x^2+y^2+1}$

$f_{xx} = \frac{(x^2+y^2+1).2 - 4x^2}{(x^2+y^2+1)^2}$

$f_{yy} = \frac{(x^2+y^2+1).2 - 4y^2}{(x^2+y^2+1)^2}$

$f_{xy} = \frac{(x^2+y^2+1).0 - 2x.2y}{(x^2+y^2+1)^2}$

$= -\frac{4xy}{(x^2+y^2+1)^2}$

Now,

$f(0,0) = 0$

$f_x(0,0) = 0$

$f_y(0,0) = 0$

$f_{xx}(0,0) = 2$

$f_{yy}(0,0) = 2$

$f_{xy}(0,0) = 0$

1st Degree Polynomial series:

$L(x,y) = f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0)$

$= 0$

(Ans.)

2nd Degree Polynomial:

$$Q(x,y) = L(x,y) + \frac{f_{xx}(0,0)}{2}(x-0)^2 + f_{xy}(0,0)\cdot(x-0)(y-0) + \frac{f_{yy}(0,0)}{2}\cdot(y-0)^2$$

$$= 0 + x^2 + 0 \cdot xy + y^2$$

$$= x^2 + y^2$$

(Ans.)

Answer to the question no. 3

Given,

$$f(x,y) = x^2 + y^2 + \frac{2}{xy}$$

We have to find all relative maxima, minima and saddle point for $f(x,y)$.

Critical Point,

$$f_x = 0$$

$$\Rightarrow 2x - \frac{2}{x^2 y} = 0$$

$$\Rightarrow y = \frac{1}{x^3} \quad \ldots (1)$$

$$f_y = 0$$

$$2y - \frac{2}{y^2 x} = 0 \quad \ldots (2)$$

Now, putting $y = \frac{1}{x^3}$ in (2) we get,

$$2\left(\frac{1}{x^3}\right) - \frac{2}{\left(\frac{1}{x^3}\right)^2 \cdot x} = 0$$

$$\Rightarrow \frac{1}{x^3} - x^5 = 0$$

$$\Rightarrow x^8 - 1 = 0$$

$$\Rightarrow (x^4)^2 - 1^2 = 0$$

$\Rightarrow (x^4+1)(x^4-1) = 0$

$\Rightarrow (x^4+1)(x^2+1)(x^2-1) = 0$

$\Rightarrow (x^2-1) = 0$ [$\because x^4+1, x^2+1$ can never be zero]

$\Rightarrow (x+1)(x-1) = 0$

$\therefore x = 1, -1$

for $x=1$,

$y = \frac{1}{x^3} = 1$

for $x=-1$

$y = \frac{1}{(-1)^3} = -1$

So, the critical points are $(1,1)$ and $(-1,-1)$

Now,

$f_{xx} = 2 - \frac{2}{y}(-2)(x^{-3})$

$= 2 + \frac{4}{x^3 y}$

$f_{yy} = 2 + \frac{4}{y^3 x}$

$f_{xy} = -\frac{2}{x^2}(-1)(y-2)$

$= \frac{2}{x^2 y^2}$

$f_{xx}(1,1) = 6$ $\quad$ $f_{xx}(-1,-1) = 6$

$f_{yy}(1,1) = 6$ $\quad$ $f_{yy}(-1,-1) = 6$

$f_{xy}(1,1) = 2$ $\quad$ $f_{xy}(-1,-1) = 2$

for $(1,1)$,

$D = f_{xx}(1,1) \cdot f_{yy}(1,1) - [f_{xy}(1,1)]^2$

$= 6 \times 6 - 4$

$= 32 > 0$ Therefore $f(x, y)$ has minimum at $(1,1)$

for $(-1,-1)$

$D = f_{xx}(-1,-1) - f_{yy}(-1,-1) - [f_{xy}(-1,-1)]^2$

$= 6 \times 6 - 4$

$= 32 > 0$, Therefore $f(x,y)$ has minimum at $(-1,-1)$ point, since $D > 0$ and $f_{xx} > 0$

Answer to the question no. 4

Given,

$$f(x,y) = xy + \frac{2}{x} + \frac{4}{y}$$

for critical points,

$f_x = 0$

$\Rightarrow y - \frac{2}{x^2} = 0$

$\Rightarrow y = \frac{2}{x^2} \cdots (1)$

$f_y = 0$

$\Rightarrow x - \frac{4}{y^2} = 0 \cdots (2)$

Now, putting $y = \frac{2}{x^2}$ in (2) we get,

$$x - \frac{4}{\frac{4}{x^4}} = 0$$

$\Rightarrow x - \frac{1}{x^4} = 0$

$\Rightarrow x^3 - 1 = 0$

$\Rightarrow (x-1)(x^2 + xy + 1) = 0$

$\therefore (x-1) = 0$

$\therefore x = 1$

putting $x=1$ at equation ①

$$y = \frac{2}{1^2}$$

$$= 2$$

So, critical point is $(1, 2)$

Now,

$$f_{xx} = -2 \cdot (-2)(x^{-3})$$

$$= \frac{4}{x^3} \quad [x=1]$$

$$= 4$$

$$f_{yy} = \frac{8}{y^3} \quad [y=2]$$

$$= 1$$

$$f_{xy} = 1$$

$$D = f_{xx}(1,2)\, f_{yy}(1,2) - [f_{xy}(1,2)]^2$$

$$= 4 \times 1 - 1^2$$

$= 3 > 0$ ∴ Therefore $D > 0$ and $f_{xx} > 0$ So, $f(x,y)$ has minimum at $(1,2)$

Answer to the question no. 5

Given,

$$\vec{F} = yze^{xy}\hat{i} + xze^{xy}\hat{j} + (e^{xy} + 3\cos 3z)\hat{k}$$

Divergence:

$$\nabla.\vec{F} = \frac{\partial}{\partial x}(yze^{xy}) + \frac{\partial}{\partial y}(xze^{xy}) + \frac{\partial}{\partial z}(e^{xy} + 3\cos 3z)$$

$$= yze^{xy}.y + xze^{xy}.x + 3(-\sin 3z)3$$

$$= ze^{xy}(x^2 + y) - 9\sin 3z$$

(Ans.)

Curl:

$$\nabla\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yze^{xy} & xze^{xy} & e^{xy} + 3\cos 3z \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial}{\partial y}(e^{xy} + 3\cos 3x) - \frac{\partial}{\partial z}(xze^{xy})\right) - \hat{j}\left(\frac{\partial}{\partial x}(e^{xy} + 3\cos 3z) - \frac{\partial}{\partial z}(yze^{xy})\right) + \hat{k}\left(\frac{\partial}{\partial x}(xze^{xy}) - \frac{\partial}{\partial y}(yze^{xy})\right)$$

$$= (xe^{xy} + 0 - xe^{xy})\hat{i} - \hat{j}((ye^{xy} + 0) - ye^{xy}) + \hat{k}(ze^{xy}.y - yze^{xy})$$

$$= 0$$

(Ans.)

Answer to the question no. 6

Given,

$$\vec{F} = (xyz)\hat{i} + y\sin z\,\hat{j} + y\cos x\,\hat{k}$$

Divergence:

$$\nabla \vec{F} = \frac{\partial}{\partial x}(xyz) + \frac{\partial}{\partial y}(y\sin z) + \frac{\partial}{\partial z}(y\cos x)$$

$$= yz + \sin z + 0$$

$$= yz + \sin z \quad \text{(Ans.)}$$

Curl:

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & y\sin z & y\cos x \end{vmatrix}$$

$$= \hat{i}\left(\frac{\partial}{\partial y} y\cos x - \frac{\partial}{\partial z} y\sin z\right) - \hat{j}\left(\frac{\partial}{\partial x} y\cos x - \frac{\partial}{\partial z} xyz\right) + \hat{k}\left(\frac{\partial}{\partial x} y\sin z - \frac{\partial}{\partial y} xyz\right)$$

$$= \hat{i}(\cos x - y\cos z) - \hat{j}(-y\sin x - xy) + \hat{k}(-xz)$$

$$= \hat{i}(\cos x - y\cos z) + \hat{j}(y\sin x + xy) - \hat{k}(xz)$$

(Ans.)